PHYS 143 – Mechanics and Thermal Physics

Reference Deck

Lecture 4: Work and Kinetic Energy. Frames moved out of the live deck. Nothing here is deleted.

Hope Donglo

More: hhdonglo.github.io/phys143.html

Equation anatomy: work by a constant force

EQUATION – work

$$W = \vec{\boldsymbol{F}} \cdot \vec{\boldsymbol{s}} = F s \cos\phi$$

Meaning. the energy a force transfers to an object as the object moves.

Variables. F force magnitude; s displacement magnitude; ϕ the angle between $\vec{\boldsymbol{F}}$ and $\vec{\boldsymbol{s}}$.

Units. $\mathrm{N\,m} = \mathrm{J}$.

Interpretation. only the part of $\vec{\boldsymbol{F}}$ along $\vec{\boldsymbol{s}}$ counts; positive speeds the object up, negative slows it, zero for a perpendicular force or no displacement.

Example. the trolley, next: $1.47 \times 10^3\,\mathrm{J}$.

Predict first

A $10\,\mathrm{N}$ force acts through $3.0\,\mathrm{m}$. Predict the work for $\phi = 0°$, $60°$, $90°$ and $180°$ before calculating.

Toolkit 1: work by a constant force

TOOLKIT 1 – Work by a constant force

1. IDENTIFY the force and the displacement of the *object it acts on*. Sketch both.
2. ANGLE: find ϕ between $\vec{\boldsymbol{F}}$ and $\vec{\boldsymbol{s}}$ (tail to tail). **3. PREDICT** the sign of W from ϕ.
4. CALCULATE: $W = Fs\cos\phi$, or $W = F_x s_x + F_y s_y$ in components (Week 1).
5. UNITS: newton $\times$ metre $=$ joule.

RULE: only the component of the force along the displacement does work.
WATCH OUT: a force perpendicular to the displacement does no work; no displacement means no work by that force.
CHECK: does the sign of the answer agree with your prediction from the sketch?

Toolkit 2: net work from the free-body diagram

TOOLKIT 2 – Net work from the free-body diagram

1. FBD: draw every force on the object (Week 3 Toolkit 1). **2. DISPLACEMENT:** draw $\vec{s}$ on the same sketch.
3. LIST each force, find ϕ for each, and assign a sign: $W > 0$, $W = 0$ or $W < 0$.
4. CALCULATE each work (Toolkit 1). **5. METHOD I:** add them with signs.
6. METHOD II (check): find the net force along $\vec{s}$ and multiply by s.

WATCH OUT: normal force and weight on a horizontal floor do no work; friction and tension may.
CHECK: do Method I and Method II agree?

Concept check: units in context

Convert before you substitute

A $4.0\,\mathrm{g}$ bullet moving at $300\,\mathrm{m/s}$ stops after penetrating $12\,\mathrm{cm}$ of wood. Find the average stopping force. A $3.0\,\mathrm{kW}$ motor runs for $2.0\,\mathrm{h}$: how much energy is that in joules?

Predict first

Which quantities are not in SI units yet? Estimate the size of the answer before calculating.

Concept check: units in context

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Predict first

Which quantities are not in SI units yet? Estimate the size of the answer before calculating.

Bullet: $m = 4.0 \times 10^{-3}\,\mathrm{kg}$, $d = 0.12\,\mathrm{m}$, $K = \frac{1}{2}mv^2 = 180\,\mathrm{J}$, so $F = K/d = 1.5 \times 10^3\,\mathrm{N}$.
Motor: $E = Pt = (3.0 \times 10^3)(7200) = 2.2 \times 10^7\,\mathrm{J}$ ($6.0\,\mathrm{kW\,h}$, with $1\,\mathrm{kW\,h} = 3.6 \times 10^6\,\mathrm{J}$).

But what if d is left in centimetres? Then $F = 180/12 = 15\,\mathrm{N}$, a hundred times too small: check the magnitude against a bullet that stops in a few centimetres.

Contact 1: what we can now do

Force and displacement

$\Downarrow$

Work $W = \vec{\boldsymbol{F}} \cdot \vec{\boldsymbol{s}} = Fs\cos\phi$

$\Downarrow$

Total work (Methods I and II)

You should be able to

- calculate work with $Fs\cos\phi$;
- decide the sign of work from the angle;
- list which forces do work on a free-body diagram;
- find the total work two ways.

Connect: Lectures 1, 5 and 11

Work is the dot product of Lecture 1 in action; the work–energy theorem of Contact 2 grows into conservation of energy (Lecture 5) and the first law (Lecture 11).

Equation anatomy: the work–energy theorem

EQUATION – work–energy theorem

$$W_{\text{tot}} = K_2 - K_1 = \Delta K$$

Meaning. the net work done on an object equals its change of kinetic energy.

Variables. W_{tot} is the work of *all* forces; K_1, K_2 are the initial and final kinetic energies.

Units. J on both sides.

Interpretation. positive net work: speeds up; negative: slows down; zero: speed unchanged.

Example. the crate: $240\,\text{J}$ takes it from 2.0 to $7.2\,\text{m/s}$.

EQUATION – kinetic energy

$$K = \tfrac{1}{2}mv^2$$

Meaning. the energy of motion; the work needed to bring the object from rest to speed v.

Variables. m mass; v speed.

Units. $\text{kg}\,\text{m}^2/\text{s}^2 = \text{J}$.

Interpretation. a scalar, never negative; $K \propto v^2$ and $K \propto m$.

Example. $1000\,\text{kg}$ at $20\,\text{m/s}$: $2.0 \times 10^5\,\text{J}$.

Toolkit 3: applying the work–energy theorem

TOOLKIT 3 – Applying the work–energy theorem

1. STATES: choose an initial point (1) and a final point (2); mark both on the sketch with the speeds.
2. FBD: list every force on the object between 1 and 2 (Week 3), and the displacement.
3. WORK: find the work of each force with its sign (Toolkits 1 and 2); add for W_{tot}.
4. WRITE $W_{\text{tot}} = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$ and solve for the unknown (a speed, a distance or a force).

WATCH OUT: the work of the net force, not of one force; forces that do no work still appear in the free-body diagram.
CHECK: does the sign of ΔK agree with whether the object speeds up or slows down?

The same problem by Newton's second law:

$$a = \frac{50 - 20}{10.0} = 3.0\,\mathrm{m/s^2}$$

$$v^2 = v_0^2 + 2a\Delta x = 4.0 + 2(3.0)(8.0) = 52$$

$$v = 7.2\,\mathrm{m/s} \quad \checkmark$$

Both methods agree. Which was shorter? Which gives the *time*?

TOOLKIT 4 – Choosing the method

ENERGY METHOD: the question is about *speed, distance or force* between two positions, and the time is not needed.
NEWTON'S SECOND LAW: the question is about *acceleration or time*, or the motion is along a path where only the net force direction matters.
WATCH OUT: energy gives no direction of motion and no time.
CHECK: solve a simple case both ways.

Contact 2: what we can now do

Total work
$\Downarrow W_{\text{tot}} = \Delta K$
Kinetic energy $K = \frac{1}{2}mv^2$
$\Downarrow$
Speed, distance or force

You should be able to

- derive and state $W_{\text{tot}} = \Delta K$;
- explain why $K \propto v^2$;
- apply the theorem to speeds, distances and forces;
- choose between the energy method and Newton's second law.

Connect: Lecture 5

$W_{\text{tot}} = \Delta K$ gives speeds without time or acceleration; potential energy next week makes it even more powerful.

Equation anatomy: work by a varying force

EQUATION – varying force

$$W = \int_{x_1}^{x_2} F_x \, \mathrm{d}x$$

Meaning. the sum of the small works $F_x \, \Delta x$ along the path.

Variables. $F_x(x)$ the force component along the displacement; x_1, x_2 start and end positions.

Units. $\mathrm{N\,m} = \mathrm{J}$.

Interpretation. the area under the F_x–x graph; area below the axis counts negative.

Example. the graph: $90\,\mathrm{J}$; the formula $3.0x^2$ from 0 to $2.0\,\mathrm{m}$: $8.0\,\mathrm{J}$.

Predict first

For $F_x = kx$ the area under the graph from 0 to X is a triangle. Predict how the work changes if X is doubled.

Toolkit 5: work from graphs and integrals

TOOLKIT 5 – Work from graphs and integrals

1. SKETCH F_x against x for the motion. **2. SPLIT** the region into rectangles and triangles.
3. AREAS: each area is a work in joules; areas below the axis are negative.
4. INTEGRATE if $F_x(x)$ is a formula: $W = \int F_x \, \mathrm{d}x$, with the limits as the start and end positions.

WATCH OUT: limits are positions x_1 and x_2, not times; check the units (N $\times$ m).
CHECK: does the integral agree with the area of a sketch?

Equation anatomy: the spring

EQUATION – Hooke's law

$$F_x = kx$$

Meaning. the force needed to hold a spring stretched or compressed by x.

Variables. k spring constant; x displacement from natural length.

Units. k in $\mathrm{N/m}$; x in m; F in N.

Interpretation. a straight line through the origin; the force by the spring is $-kx$, restoring.

Example. the bathroom scale: $k = 7.4 \times 10^4\,\mathrm{N/m}$.

EQUATION – work on a spring

$$W = \tfrac{1}{2}kx_2^2 - \tfrac{1}{2}kx_1^2$$

Meaning. the work to stretch a spring from x_1 to x_2 (the area under the line).

Variables. k; x_1, x_2 measured from natural length.

Units. $\mathrm{N/m} \times \mathrm{m}^2 = \mathrm{J}$.

Interpretation. quadratic: doubling the stretch quadruples the work.

Example. the glider: $\tfrac{1}{2}(40.0)(0.15)^2 = 0.45\,\mathrm{J}$.

TOOLKIT 6 – Springs

1. **NATURAL LENGTH:** measure x from the unstretched (or uncompressed) position.
2. **DIRECTION:** the spring's force on the object points back toward $x = 0$: $F = -kx$.
3. **WORK:** $W_{\text{by spring}} = \frac{1}{2}kx_1^2 - \frac{1}{2}kx_2^2$ between x_1 and x_2; the work you do on the spring is the negative.
4. **COMBINE** with other forces using the work–energy theorem (Toolkit 3).

WATCH OUT: use the stretch from natural length, not the spring's total length; signs of the work by and on the spring.

CHECK: is k in N/m and x in m?

Concept check: work by tension on connected masses

Decide first

Blocks of $3.0\,\mathrm{kg}$ and $5.0\,\mathrm{kg}$ hang from a light rope over a frictionless, massless pulley and are released from rest. The heavier block descends $2.0\,\mathrm{m}$. What is the sign of the work done by the tension on each block, and the total work by the rope?

Predict first

The tension is the same on both blocks. Compare its direction with each displacement.

Concept check: work by tension on connected masses

Decide first

Blocks of $3.0\,\mathrm{kg}$ and $5.0\,\mathrm{kg}$ hang from a light rope over a frictionless, massless pulley and are released from rest. The heavier block descends $2.0\,\mathrm{m}$. What is the sign of the work done by the tension on each block, and the total work by the rope?

Predict first

The tension is the same on both blocks. Compare its direction with each displacement.

$T = \dfrac{2m_1m_2}{m_1+m_2}g = 36.75\,\mathrm{N}$. On the rising $3.0\,\mathrm{kg}$ block $W_T = +Ts = +73.5\,\mathrm{J}$; on the falling $5.0\,\mathrm{kg}$ block $W_T = -73.5\,\mathrm{J}$. The rope does **no net work**: it only transfers energy. Net work: $98.0 - 73.5 = 24.5\,\mathrm{J}$ on the $5.0\,\mathrm{kg}$ block, $73.5 - 58.8 = 14.7\,\mathrm{J}$ on the $3.0\,\mathrm{kg}$ block; total $39.2\,\mathrm{J} = (m_2 - m_1)gs$. ✓

But what if the pulley has mass or friction? The tensions on the two sides then differ, and the pulley absorbs or stores energy.

Equation anatomy: power

EQUATION – power

$$P_{\text{av}} = \frac{\Delta W}{\Delta t}, \qquad P = \frac{\mathrm{d}W}{\mathrm{d}t} = \vec{\boldsymbol{F}} \cdot \vec{\boldsymbol{v}}$$

Meaning. the rate at which a force transfers energy to an object.

Variables. W work; t time; $\vec{\boldsymbol{F}}$ force; $\vec{\boldsymbol{v}}$ velocity of its point of application.

Units. $\mathrm{J/s} = \mathrm{W}$ (watt).

Interpretation. the same work done faster means more power; $P = Fv\cos\phi$ at one instant.

Example. the climber: $4.7 \times 10^2\,\mathrm{W}$; the car: $15\,\mathrm{kW}$.

Predict first

Two climbers lift the same load by the same height; one takes twice as long.
Compare the work and the power.

Toolkit 7: power

TOOLKIT 7 – Power

1. **AVERAGE POWER:** $P_{\text{av}} = W/\Delta t$ (total work over the time taken).
2. **INSTANTANEOUS POWER:** $P = \vec{\boldsymbol{F}} \cdot \vec{\boldsymbol{v}} = Fv\cos\phi$ at one instant.
3. **UNITS:** watt, kilowatt, horsepower; convert carefully.
4. **COMPARE:** to find a time, use $t = W/P$; to find a force at constant P, use $F = P/v$.

WATCH OUT: power depends on how fast the work is done, not on the amount of work alone.

CHECK: is the power of a human of order $10^2\,\text{W}$, a car of order $10^4\,\text{W}$?

Connect: where work and power are used next

Lecture 5

The work done by gravity and by springs is path-independent, so it can be stored as potential energy; $W_{\text{tot}} = \Delta K$ becomes conservation of energy.

Lectures 9 to 11

The kinetic energy removed by friction appears as thermal energy; the work done by a gas is $\int p\,\mathrm{d}V$, the same integral with $F\,\mathrm{d}x$.

Lecture 8

Rotational work $W = \tau\,\Delta\theta$ and power $P = \tau\omega$ mirror $W = Fs$ and $P = Fv$.

Physical meaning

Energy is the central bookkeeping idea of the course: the same $W_{\text{tot}} = \Delta K$ method returns in every remaining lecture.

Master procedure: work and energy problems

MASTER PROCEDURE – ENERGY

1. IDENTIFY The object, and what is asked: a speed, a distance, a force, a work or a power.
2. STATES Mark the initial and final positions with their speeds.
3. DRAW THE FBD All forces on the object, and the displacement (Week 3).
4. WORK OF EACH FORCE Sign first, then size: constant, graph, integral or spring (Toolkits 1 to 6).
5. PREDICT Sign of W_{tot}, speeding up or slowing down, rough size.
6. SELECT THE METHOD Energy method or Newton's second law (Toolkit 4).
7. EXECUTE $W_{\text{tot}} = \Delta K$; add power if a rate is asked (Toolkit 7).
8. EVALUATE Units, sign of ΔK, check by a second method.

SKETCH $\rightarrow$ PREDICT $\rightarrow$ CALCULATE $\rightarrow$ CHECK.

Inside the course framework

IDENTIFY $\leftrightarrow$ steps 1–2
SET UP $\leftrightarrow$ steps 3–6
EXECUTE $\leftrightarrow$ step 7
EVALUATE $\leftrightarrow$ step 8

Physical meaning

Next lecture: potential energy turns the work done by gravity and springs into stored energy, and leads to conservation of energy.

The Physicist's Toolkit: Week 4 index

THE SEVEN TOOLKITS

1	**Work by a constant force**	force and displacement; angle ϕ; sign; $W = Fs\cos\phi$
2	**Net work from the FBD**	list forces; sign each; Method I and Method II
3	**The work–energy theorem**	two states; forces doing work; $W_{\text{tot}} = \Delta K$
4	**Choosing the method**	energy for speed and distance; Newton for acceleration and time
5	**Work from graphs and integrals**	area under F_x–x; $\int F_x \, \mathrm{d}x$
6	**Springs**	Hooke's law; $W = \frac{1}{2}kx^2$; signs by and on the spring
7	**Power**	$W/\Delta t$ and $\vec{\boldsymbol{F}} \cdot \vec{\boldsymbol{v}}$; watt, kW, hp

The Master Procedure sits above these. Week 1 to 3 toolkits (dot product, components, free-body diagrams, friction) are reused, not renamed.

Summary: key ideas

Key ideas

- Only the part of the force along the displacement does work.
- A perpendicular force, or no displacement, means no work.
- Net work changes kinetic energy, and so the speed.
- Energy methods give speeds and distances without time.
- Work is the area under the F_x–x graph.
- Spring work is quadratic: $\frac{1}{2}kx^2$.

Where to look

Definitions: the vocabulary and equation-anatomy slides of each contact.

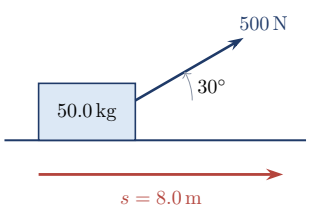
Common mistakes: the last slide of each contact.

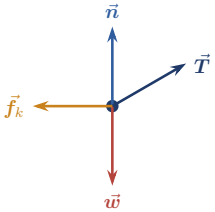
Connections: the Connect boxes and slides.

Summary: core equations

Idea	Equation	Conditions and units
Work, constant force	$W = \vec{\boldsymbol{F}} \cdot \vec{\boldsymbol{s}} = Fs\cos\phi$	constant $\vec{\boldsymbol{F}}$; J
Total work	$W_{\text{tot}} = \sum W_i = (\sum \vec{\boldsymbol{F}}) \cdot \vec{\boldsymbol{s}}$	signed sum
Kinetic energy	$K = \frac{1}{2}mv^2$	scalar, ≥ 0; J
Work–energy theorem	$W_{\text{tot}} = \Delta K$	all forces; particle
Work by gravity	$W_w = \pm mgh$	down $+$, up $-$
Varying force	$W = \int_{x_1}^{x_2} F_x \, \mathrm{d}x$	area under F_x–x
Spring	$F_x = kx, \quad W = \frac{1}{2}kx_2^2 - \frac{1}{2}kx_1^2$	x from natural length; N/m
Power	$P_{\text{av}} = W/\Delta t, \quad P = \vec{\boldsymbol{F}} \cdot \vec{\boldsymbol{v}}$	W

Example 2: a tractor drags a crate





Example 2

A tractor drags a $50.0\,\mathrm{kg}$ crate of yams $8.0\,\mathrm{m}$ across level ground. The cable force is $500\,\mathrm{N}$, 30° above the horizontal. The coefficient of kinetic friction is $\mu_k = 0.20$. Find the work done by each force and the total work.

Key idea

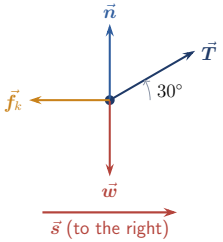
Draw the free-body diagram first. n is not mg because $\vec{T}$ has a vertical part.

Predict first

Which forces do positive work, negative work, no work? Is the total positive?

Example 2: a tractor drags a crate, solution (1)

CORE



Key idea

Friction needs n, and n needs the vertical equation.

IDENTIFY

Given: $m = 50.0\,\text{kg}$, $T = 500\,\text{N}$ at 30°, $s = 8.0\,\text{m}$, $\mu_k = 0.20$. Find: W of each force and W_{tot}.

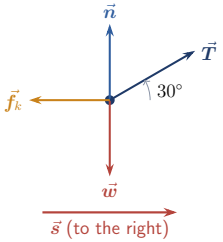
SET UP

Forces: $\vec{\boldsymbol{T}}$, $\vec{\boldsymbol{n}}$, $\vec{\boldsymbol{w}}$, $\vec{\boldsymbol{f}}_k$. Work of each: (1), $W = \vec{\boldsymbol{F}} \cdot \vec{\boldsymbol{s}}$. Total: (3).

Friction first needs n (Week 3). The crate does not move vertically: $\vec{\boldsymbol{n}} + \vec{\boldsymbol{w}} + \vec{\boldsymbol{T}} + \vec{\boldsymbol{f}}_k$ has $\sum F_y = 0$.

Example 2: a tractor drags a crate, solution (1)

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EXECUTE: normal force and friction

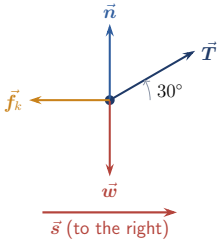
Vertical equation: $n + T\sin 30^\circ - mg = 0$

Solve: $n = mg - T\sin 30^\circ =$
$(50.0)(9.8)\,\text{N} - (500)(0.500)\,\text{N} = 490\,\text{N} - 250\,\text{N} = 240\,\text{N}$

Friction: $f_k = \mu_k n = (0.20)(240\,\text{N}) = 48.0\,\text{N}$

Example 2: a tractor drags a crate, solution (1)

CORE



Key idea

Friction needs n, and n needs the vertical equation.

IDENTIFY

Given: $m = 50.0\,\text{kg}$, $T = 500\,\text{N}$ at 30°, $s = 8.0\,\text{m}$, $\mu_k = 0.20$. Find: W of each force and W_{tot}.

SET UP

Forces: $\vec{\boldsymbol{T}}$, $\vec{\boldsymbol{n}}$, $\vec{\boldsymbol{w}}$, $\vec{\boldsymbol{f}}_k$. Work of each: (1), $W = \vec{\boldsymbol{F}} \cdot \vec{\boldsymbol{s}}$. Total: (3).

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EXECUTE: normal force and friction

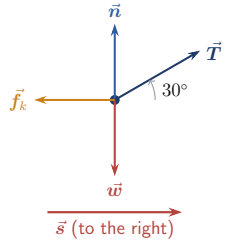
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Friction: $f_k = \mu_k n = (0.20)(240\,\text{N}) = 48.0\,\text{N}$

Next frame: the work of each force, Methods I and II, and EVALUATE.

Example 2: a tractor drags a crate, solution (2)



EXECUTE: Method I, add the works

Cable: $W_T = Ts\cos 30^\circ = (500\,\text{N})(8.0\,\text{m})(0.8660) = +3.46 \times 10^3\,\text{J}$

Friction ($\phi = 180^\circ$):

$W_f = -f_k s = -(48.0\,\text{N})(8.0\,\text{m}) = -384\,\text{J}$

Weight and normal force ($\phi = 90^\circ$): $W_w = W_n = 0$

Total: $W_{\text{tot}} = 3464\,\text{J} - 384\,\text{J} = 3080\,\text{J}$

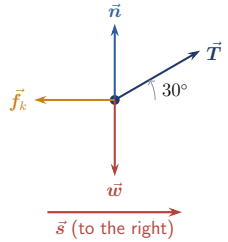
$$\boxed{W_{\text{tot}} = 3.08 \times 10^3\,\text{J}}$$

Key idea

Friction takes away part of what the cable gives: W_{tot} is smaller than W_T.

Example 2: a tractor drags a crate, solution (2)

CORE



Key idea

Friction takes away part of what the cable gives: W_{tot} is smaller than W_T.

EXECUTE: Method I, add the works

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Friction ($\phi = 180^\circ$):

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Total: $W_{\text{tot}} = 3464\,\text{J} - 384\,\text{J} = 3080\,\text{J}$

$$\boxed{W_{\text{tot}} = 3.08 \times 10^3\,\text{J}}$$

EXECUTE: Method II, check with the net force

Equation (3): $W_{\text{tot}} = (\sum \vec{F}) \cdot \vec{s}$, with

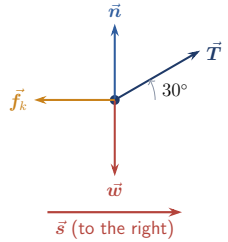
$\sum F_x = T\cos 30^\circ - f_k$

Substitute: $\sum F_x = 433.0\,\text{N} - 48.0\,\text{N} = 385.0\,\text{N}$, so

$W_{\text{tot}} = (385.0\,\text{N})(8.0\,\text{m}) = 3080\,\text{J}$

Example 2: a tractor drags a crate, solution (2)

CORE



Key idea

Friction takes away part of what the cable gives: W_{tot} is smaller than W_T.

EXECUTE: Method I, add the works

Cable: $W_T = Ts\cos 30^\circ = (500\,\text{N})(8.0\,\text{m})(0.8660) = +3.46 \times 10^3\,\text{J}$

Friction ($\phi = 180^\circ$):

$W_f = -f_k s = -(48.0\,\text{N})(8.0\,\text{m}) = -384\,\text{J}$

Weight and normal force ($\phi = 90^\circ$): $W_w = W_n = 0$

Total: $W_{\text{tot}} = 3464\,\text{J} - 384\,\text{J} = 3080\,\text{J}$

$$\boxed{W_{\text{tot}} = 3.08 \times 10^3\,\text{J}}$$

EXECUTE: Method II, check with the net force

Equation (3): $W_{\text{tot}} = (\sum \vec{F}) \cdot \vec{s}$, with

$\sum F_x = T\cos 30^\circ - f_k$

Substitute: $\sum F_x = 433.0\,\text{N} - 48.0\,\text{N} = 385.0\,\text{N}$, so

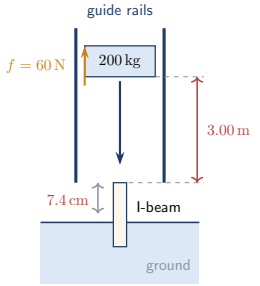
$W_{\text{tot}} = (385.0\,\text{N})(8.0\,\text{m}) = 3080\,\text{J}$

EVALUATE

Signs as predicted: $W_T > 0$, $W_f < 0$, $W_w = W_n = 0$.

The two methods agree. Units: J. ✓

Example 5: a pile driver



Example 5

At a building site a $200\,\mathrm{kg}$ hammerhead is released from rest $3.0\,\mathrm{m}$ above a post. It falls freely onto the post and drives it $0.050\,\mathrm{m}$ into the ground. Find (a) the hammer's speed at impact, and (b) the average force the post exerts on the hammer while it is being driven in.

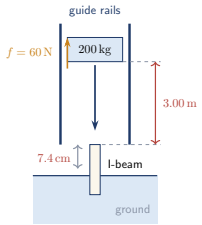
Key idea

Two stages: free fall (gravity does the work), then driving the post in (the post's resistive force does negative work).

Predict first

Is the average force larger or smaller than the hammer's weight, $1960\,\mathrm{N}$? By roughly how many times?

Example 5(a): the fall



Key idea

The mass cancels: $v = \sqrt{2gh}$, as in free fall (Week 2).

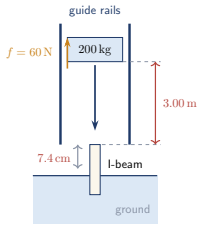
IDENTIFY

Given: $m = 200\,\mathrm{kg}$, $h = 3.0\,\mathrm{m}$, $v_1 = 0$. Find: v_2 just before impact.

SET UP

State 1: released. State 2: just before impact. Only gravity acts, moving down: use (6), $W_w = +mgh$, and (5).

Example 5(a): the fall



Key idea

The mass cancels: $v = \sqrt{2gh}$, as in free fall (Week 2).

IDENTIFY

Given: $m = 200\,\mathrm{kg}$, $h = 3.0\,\mathrm{m}$, $v_1 = 0$. Find: v_2 just before impact.

SET UP

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EXECUTE

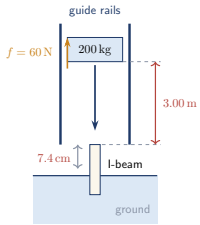
Gravity: $W_w = mgh = (200\,\mathrm{kg})(9.8\,\mathrm{m/s^2})(3.0\,\mathrm{m}) = 5880\,\mathrm{J}$

Theorem: $mgh = \frac{1}{2}mv_2^2 - 0$, so

$v_2 = \sqrt{2gh} = \sqrt{2(9.8)(3.0)}\,\mathrm{m/s}$

$\boxed{v_2 = 7.7\,\mathrm{m/s}}, \quad K_2 = 5880\,\mathrm{J}$

Example 5(a): the fall



Key idea

The mass cancels: $v = \sqrt{2gh}$, as in free fall (Week 2).

IDENTIFY

Given: $m = 200\,\mathrm{kg}$, $h = 3.0\,\mathrm{m}$, $v_1 = 0$. Find: v_2 just before impact.

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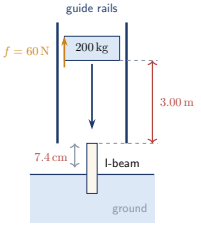
$v_2 = \sqrt{2gh} = \sqrt{2(9.8)(3.0)}\,\mathrm{m/s}$

$\boxed{v_2 = 7.7\,\mathrm{m/s}}$, $\quad K_2 = 5880\,\mathrm{J}$

EVALUATE

Matches free fall from Week 2, $v^2 = 2gh$, and the mass cancelled. ✓

Example 5(b): driving the post



Key idea

The post stops $5880\,\mathrm{J}$ of motion in $0.050\,\mathrm{m}$: a large force.

IDENTIFY

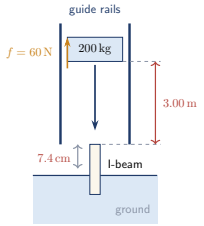
Given: $m = 200\,\mathrm{kg}$, $d = 0.050\,\mathrm{m}$, $K_1 = 5880\,\mathrm{J}$ (at contact), $K_2 = 0$. Find: the average resistive force R of the post on the hammer.

SET UP

(5) from contact to rest. Weight acts along $\vec{s}$: $W_w = +mgd$. The resistive force R acts against $\vec{s}$: $W_R = -Rd$.

Example 5(b): driving the post

CORE



Key idea

The post stops $5880\,\mathrm{J}$ of motion in $0.050\,\mathrm{m}$: a large force.

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Given: $m = 200\,\mathrm{kg}$, $d = 0.050\,\mathrm{m}$, $K_1 = 5880\,\mathrm{J}$ (at contact), $K_2 = 0$. Find: the average resistive force R of the post on the hammer.

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EXECUTE

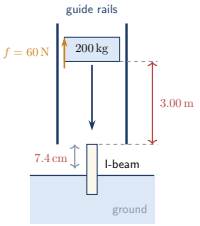
Equation: $W_{\text{tot}} = mgd - Rd = 0 - K_1$

Solve:

$$R = mg + \frac{K_1}{d} = 1960\,\mathrm{N} + \frac{5880\,\mathrm{J}}{0.050\,\mathrm{m}} = 1960\,\mathrm{N} + 117\,600\,\mathrm{N}$$

$$\boxed{R = 1.2 \times 10^5\,\mathrm{N}}$$

Example 5(b): driving the post



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The post stops $5880\,\mathrm{J}$ of motion in $0.050\,\mathrm{m}$: a large force.

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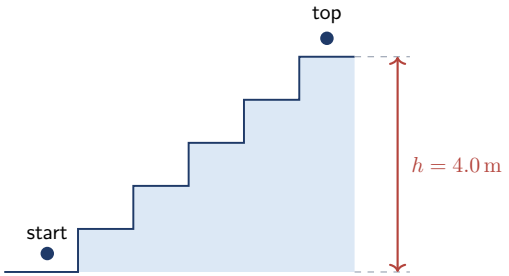
$$R = mg + \frac{K_1}{d} = 1960\,\mathrm{N} + \frac{5880\,\mathrm{J}}{0.050\,\mathrm{m}} = 1960\,\mathrm{N} + 117\,600\,\mathrm{N}$$

$$\boxed{R = 1.2 \times 10^5\,\mathrm{N}}$$

EVALUATE

About 61 times the weight, as predicted. Check over the whole motion: $mg(h + d) = Rd$ gives

$R = \dfrac{1960(3.050)}{0.050} = 1.2 \times 10^5\,\mathrm{N}$. ✓By Newton's third law the hammer pushes the post down with the same force.



Example 10

A $60\,\mathrm{kg}$ student climbs $4.0\,\mathrm{m}$ of stairs in $5.0\,\mathrm{s}$. (a) Find the average power. (b) An $80\,\mathrm{kg}$ student develops the same power. How long does the same climb take?

Predict first

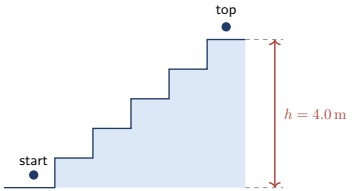
Will the heavier student take longer or less time? About how large should the power be: 1, 10^2 or $10^4\,\mathrm{W}$?

Key idea

At constant speed you do work mgh against gravity. Power is that work over the time.

Example 10: two stair climbers, solution

CORE



Key idea

Same power, more weight: the work is larger, so the time is longer.

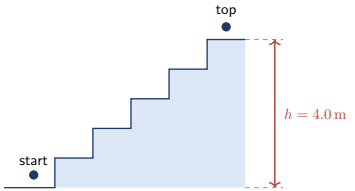
IDENTIFY

Given: $m_1 = 60\,\mathrm{kg}$, $h = 4.0\,\mathrm{m}$, $t_1 = 5.0\,\mathrm{s}$; $m_2 = 80\,\mathrm{kg}$, same P. Find: P_{av}, t_2.

SET UP

Constant speed, so the work is mgh (6). Average power (10): $P_{\mathrm{av}} = W/\Delta t$.

Example 10: two stair climbers, solution



Key idea

Same power, more weight: the work is larger, so the time is longer.

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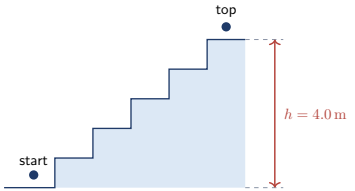
EXECUTE

(a) Work: $W_1 = mgh = (60\,\mathrm{kg})(9.8\,\mathrm{m/s^2})(4.0\,\mathrm{m}) = 2350\,\mathrm{J}$

Power: $P_{\mathrm{av}} = W_1/t_1 = 2350\,\mathrm{J}/5.0\,\mathrm{s}$ $\boxed{P_{\mathrm{av}} = 4.7 \times 10^2\,\mathrm{W}}$

(b) Work: $W_2 = (80)(9.8)(4.0)\,\mathrm{J} = 3136\,\mathrm{J}$

Time: $t_2 = W_2/P_{\mathrm{av}} = 3136\,\mathrm{J}/470\,\mathrm{W}$ $\boxed{t_2 = 6.7\,\mathrm{s}}$



Key idea

Same power, more weight: the work is larger, so the time is longer.

IDENTIFY

Given: $m_1 = 60\,\mathrm{kg}$, $h = 4.0\,\mathrm{m}$, $t_1 = 5.0\,\mathrm{s}$; $m_2 = 80\,\mathrm{kg}$, same P. Find: P_{av}, t_2.

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Time: $t_2 = W_2/P_{\mathrm{av}} = 3136\,\mathrm{J}/470\,\mathrm{W}$ $\boxed{t_2 = 6.7\,\mathrm{s}}$

EVALUATE

Heavier means slower, as predicted: $t_2/t_1 = m_2/m_1 = 1.33$.

About $10^2\,\mathrm{W}$ is right for a person. ✓

What K means

From Example 5(a): K is the work done to bring an object **from rest to its speed**. From 5(b): K is the work the object can do in being **brought to rest**.

Predict first

Which has more kinetic energy: a $1000\,\mathrm{kg}$ taxi at $20\,\mathrm{m/s}$, or a $2000\,\mathrm{kg}$ trotro at $10\,\mathrm{m/s}$?

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Taxi: $K = \frac{1}{2}(1000)(20)^2 = 2.0 \times 10^5\,\mathrm{J}$. Trotro: $K = \frac{1}{2}(2000)(10)^2 = 1.0 \times 10^5\,\mathrm{J}$. The taxi has **twice** the kinetic energy, although the trotro has twice the mass.

Two canoes, equal K

Two fishing canoes finish a race with equal kinetic energy. One has twice the mass of the other. What is the ratio of their speeds?

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Two fishing canoes finish a race with equal kinetic energy. One has twice the mass of the other. What is the ratio of their speeds?

$\frac{1}{2}m_1v_1^2 = \frac{1}{2}m_2v_2^2$ with $m_1 = 2m_2$ gives $v_2/v_1 = \sqrt{2}$. The lighter canoe moves $\sqrt{2}$ times faster.

Common error

Doubling the speed multiplies K by 4, not 2. Doubling the mass doubles K.